

Identity-Consistent Analysis of Long-Shot Windsurfing Video

An Offline System for Tracking and Rider-Relative Stabilization

Bertil Braun

contact@bertil-braun.de

Abstract

Long-shot windsurfing video combines small targets, large camera pans, prolonged overlaps, and rapidly changing backgrounds. The desired output is not a generic MOT trace but a separate, stable rider-relative video for each surfer; one false identity merge can invalidate an otherwise useful result. We present an offline analysis system that detects surfers, forms conservative local tracklets, links them globally with camera-compensated motion and a foreground-masked sail-color descriptor, and uses two rig keypoints to drive a stable virtual camera. The tracking stage is evaluated on 21 manually reconstructed development videos containing 41,004 retained observations. On this fixed-observation protocol, the production system achieves 0.957 pairwise precision, 0.918 recall, and 0.937 F1, compared with 0.792 F1 for OC-SORT and 0.828 for BoT-SORT. It reduces fragmentation excess to 42, from 845 and 244 respectively, while retaining nine contaminated tracks. Ablations and a visual audit show that conservative geometric vetoes remove many catastrophic false links; the remaining failures concentrate at crossings and visually near-duplicate sails. The result is an applied computer-vision system report: tracking receives comparative quantitative evaluation, while pose-based stabilization is documented as the downstream system output and evaluated qualitatively.

1 Introduction

Shore-recorded windsurfing video is visually rich but difficult to repurpose into a focused view of one athlete. The camera follows the action with large pans, riders occupy few pixels, rigs overlap for extended periods, and an athlete may disappear behind another sail before returning against a different background. A useful system must do more than draw plausible boxes: it must preserve one physical identity long enough to produce a stable rider-relative video.

This requirement changes the engineering objective. A missed association leaves two fragments that can be reviewed or joined later. A false association silently inserts another surfer into the output and can invalidate the complete sequence. Standard online trackers are designed for causal decisions and balanced benchmark metrics; this project instead has the complete recording available and assigns a much larger cost to false merging than to fragmentation.

We describe the resulting end-to-end computer-vision

system. A manually trained detector supplies surfer boxes and two rig landmarks. Masked global-motion estimation separates camera pan from target motion. Conservative online association creates high-purity tracklets, after which an offline integer program links compatible tracklets using motion, temporal gap, and a sail-specific color representation. Finally, the rig landmarks define a temporally smoothed anchor and scale for a stable virtual camera.

The contributions are:

- an explicit formulation of long-shot windsurfing analysis around asymmetric identity failure cost and complete-video processing;
- a domain-specific association pipeline combining foreground-masked sail appearance, camera-compensated four-dimensional motion, conservative tracklets, geometric vetoes, and global path optimization;
- a two-landmark rider-relative stabilization signal designed around windsurf rig geometry rather than generic human pose; and
- a fixed-observation evaluation on 21 manually reconstructed videos, including OC-SORT and BoT-SORT comparisons, staged ablations, and a link-level failure audit.

The document is best read as an applied systems paper and technical report. The tracking subsystem has quantitative comparative evidence. Detection supplies the observation boundary, and pose stabilization demonstrates the downstream system contract, but neither is presented as a general benchmark contribution.

2 Problem Regime and Requirements

The input is a complete windsurfing recording captured from shore. A scene usually contains only a handful of surfers, but each target can be small and several sails may remain close or overlap for seconds. The camera follows the action with pans of about 90° in demanding examples, sometimes more. Consequently almost every background feature moves in image coordinates, and newly revealed portions of the scene may share no features with earlier frames.

The desired result is one continuous, dense trajectory per physical surfer from which a focused video can be rendered. This gives three requirements.

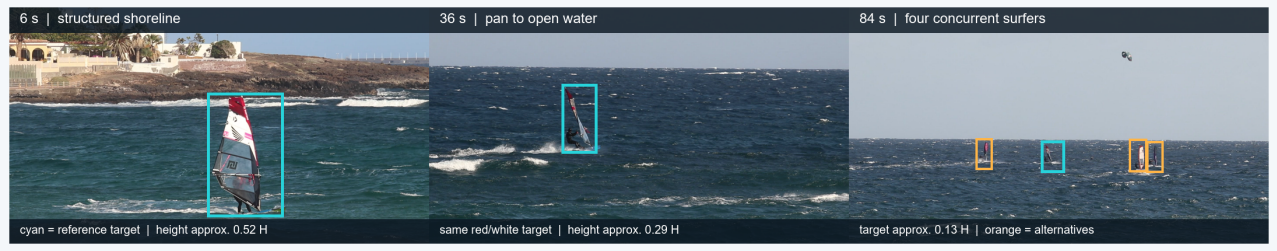


Figure 1: Operating regime in one recording at 6, 36, and 84s. Cyan follows the same red-and-white sail while the background and camera direction change; the target height drops from approximately $0.52H$ to $0.13H$. Orange marks plausible competing surfers. The example combines scale change, camera motion, and identity ambiguity within one continuous shot.

Identity is the primary invariant. A track that switches rider once is not partly correct from the user’s perspective: the extracted sequence is contaminated. Localization smoothness and recall matter only after identity is preserved.

Association errors are asymmetric. A missed link creates fragments that remain inspectable. A false link combines two identities and is difficult to detect downstream. The system should therefore abstain when evidence is insufficient. “Near-zero false merges” is the design target, not a claim that the implementation achieves perfect tracking.

The complete recording is available. There is no real-time constraint. Causal processing is useful for forming local tracklets, but ambiguous choices need not be finalized before later evidence arrives. This motivates a two-stage design: conservative local association followed by global offline linking.

Large camera motion makes raw image displacement a poor association cue. The common background motion must first be estimated and applied to the state prediction. Long gaps remain intrinsically ambiguous: a later fragment may be a reappearance or a different surfer entering after the earlier track ended. Appearance is therefore required alongside motion, but the representation should describe the visible sail rather than a generic pedestrian.

3 Related Work

Online tracking by detection. SORT combines a Kalman filter with frame-wise assignment and established a strong, minimal online baseline [1]. Deep SORT adds a learned pedestrian appearance metric to survive longer occlusions [2], while ByteTrack improves continuity by associating low-confidence detections rather than discarding them immediately [3]. OC-SORT revisits motion estimation after occlusion and uses observations to correct accumulated prediction error [4]. BoT-SORT combines motion, camera compensation, and optional appearance features in a stronger pedestrian tracker [5]; Deep OC-SORT further adapts the contribution of appearance under feature degradation [6]. These systems motivate our motion and appearance cues, but remain fundamentally causal at association time.

Offline and global association. Batch tracking has long been formulated as network flow or related graph optimization [7]. More recent work learns objectives for multi-frame flow association rather than training only local pair costs [8]. Our integer program belongs to this family but targets a small, bounded graph of already conservative tracklets. It is intentionally simple: start, end, discard, and link choices are optimized jointly, while the domain-specific work lies in constructing reliable fragments and costs.

Motion, appearance, and evaluation. Kalman filtering [10] and Rauch–Tung–Striebel smoothing [11] provide the temporal state machinery used before and after association. Learned ReID is effective when its training semantics match the target class, but pedestrian embeddings allocate capacity to body and clothing cues that occupy very few pixels in our footage. We instead exploit the sail: a large, saturated, deliberately distinctive surface. Standard MOT metrics such as HOTA explicitly balance detection, association, and localization [9]. Because our saved observations are fixed and the product loss is dominated by identity contamination, we report pair-clustering precision and recall together with absolute contaminated-track and fragmentation counts rather than claiming an end-to-end MOT score.

Pose-guided framing. The stabilization stage is related to pose-guided cropping, but its goal is not human pose recovery or action recognition. A windsurf rig is a large articulated object whose axis-aligned detector box changes strongly with rotation. Two semantic rig points are sufficient to construct a rider-relative anchor and scale, making the pose representation deliberately task-specific.

4 System Overview

The pipeline separates perception, identity, and presentation geometry:

$$\begin{aligned}
 \text{video} &\rightarrow \text{boxes} + \text{landmarks} \\
 &\rightarrow \text{local tracklets} \\
 &\rightarrow \text{offline identity paths} \\
 &\rightarrow \text{stable rider views.}
 \end{aligned} \tag{1}$$

The detector is a one-class YOLO pose model trained from manually reviewed boxes and two windsurf-specific

landmarks: the boom–mast junction and mast tip. Every box is also converted into a compact color descriptor. Independently, background features are estimated between adjacent frames while masking detected surfers; the resulting camera transforms are passed to both tracking stages.

The first association stage is causal but deliberately conservative. It accepts only locally unambiguous matches and exposes uncertainty as separate tracklets. The second stage sees the complete recording, creates a bounded graph of feasible tracklet continuations, and solves for disjoint paths. Post-processing smooths linked boxes and interpolates internal gaps, but retains the distinction between observed and synthesized detections. The pose stage then converts every dense identity track into a frame-indexed virtual-camera signal.

The components have explicit contracts. Detection proposes observations; local association should prioritize fragment purity; global association should prefer a split over implausible identity evidence; and stabilization must never change track identity. This modularity is important because only the association contract is evaluated against alternative trackers, while the final view also depends on detector and landmark quality.

5 Offline Identity Association

The tracker converts detections into high-purity local fragments and then into globally consistent paths. The same camera-motion estimates support both stages.

5.1 Foreground-masked sail appearance

Windsurfing sails are large, saturated, and often visually distinctive even when the rider occupies only a few pixels. For each detection crop we build a descriptor from a joint histogram of the chromatic Lab channels (a, b), a circular hue histogram, and a lower-weight luminance histogram. Saturation weighting emphasizes colored sail panels over grey water, spray, and haze. Three vertical bands preserve coarse color layout, and a fourth block represents the complete crop. Each block is L1-normalized and square-root mapped before the concatenated vector is L2-normalized, yielding a bounded Hellinger-style distance.

The descriptor first removes likely background using a border-derived Lab model. A smoothness gate identifies water-like pixels, and only background components connected to the crop border are removed. Morphological cleanup and a minimum-mask fallback avoid empty descriptors. This foreground mask is important: an earlier inactive code path used a fixed centre rectangle, allowing water and overlapping sails to dilute the intended cue.

For a tracklet, all observation embeddings are averaged into a whole-fragment prototype. This replaced endpoint-biased exponential averages, which emphasized the last crop of the earlier fragment and first crop of the later fragment—exactly the frames most often contaminated during a crossing. Given descriptor dis-



Figure 2: Production descriptor on consecutive crops of two sails. Repeated observations receive high match scores, while the cross-sail entries reach the lower clip bound. The hand-selected example explains the cue; it is not an accuracy result.

tance d , the current heuristic match score is

$$p_a = \text{clip}_{[10^{-6}, 1-10^{-6}]}(1 - 11.630d). \quad (2)$$

It is a tuned association score, not a calibrated posterior probability.

5.2 Conservative local tracklets

Each detection initially forms a singleton. A greedy preprocessor advances a constant-velocity Kalman state [10] through the estimated frame-to-frame camera transforms and compares the prediction with current detections. Strict links require both motion and appearance evidence; looser links are accepted only when they are unique. Duplicate proposals, competing successors, or multiple plausible candidates remain separate. The local stage therefore sacrifices completeness to preserve fragment purity.

This behavior is deliberate. In the evaluation reconstruction, 598 of 601 preprocessor fragments are pure with respect to retained gold identities. The dominant catastrophic error is introduced later when two pure fragments are linked incorrectly, which makes the offline edge decision the appropriate place for additional safeguards.

5.3 Candidate graph

For non-overlapping fragments T_i and T_j , a directed candidate $i \rightarrow j$ is created only when the gap is at most three seconds. A Kalman state fitted to all detections of T_i is propagated to the first observations of T_j , applying global camera transforms at every intervening frame. Unlike the earlier production call, the gate uses the complete measurement (c_x, c_y, w, h) rather than discarding width and height.

Every candidate receives

$$\begin{aligned} C_{ij} &= w_m C_{ij}^{\text{motion}} + w_a C_{ij}^{\text{appearance}} \\ &\quad + w_g C_{ij}^{\text{gap}}, \\ C_{ij}^{\text{gap}} &= -\Delta f_{ij} \log p_{\text{miss}}. \end{aligned} \quad (3)$$

Motion is the mean negative log score from the first two observations of T_j ; appearance uses the whole-fragment

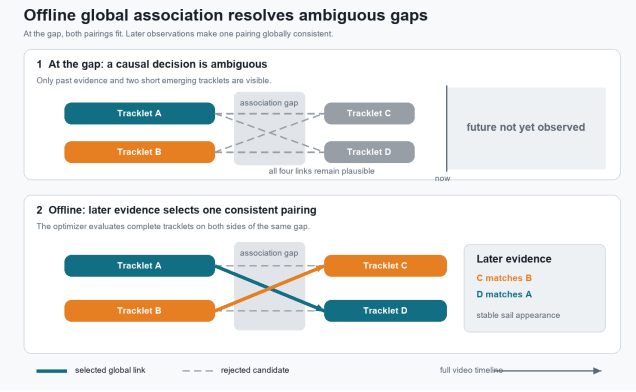


Figure 3: Complete-video evidence supports global tracklet association. Later stable observations distinguish locally ambiguous successors; the optimizer then chooses a compatible pairing jointly. Start, end, discard, and gap costs are omitted from the conceptual diagram.

prototypes; and the gap term increases with the number of missing frames.

The graph now applies absolute plausibility vetoes before optimization. Edges are removed when motion NLL exceeds 8, or when endpoint box area or aspect ratio changes by more than a factor of three. The Kalman filter already models size; the explicit ratios supplement it with an interpretable discontinuity bound. These rules implement abstention at the evidence boundary. They are necessary because a central fragment’s start and end penalties can otherwise make even poor finite evidence cheaper than leaving two trajectories separate.

5.4 Global path optimization

Binary variables represent starting or ending a path at each fragment, selecting a candidate edge, or discarding an eligible short fragment. Each retained fragment has exactly one incoming choice and one outgoing choice, so selected links form disjoint, time-ordered paths. Start and end costs are reduced near image borders, where a surfer entering or leaving the field of view is more plausible. An edge that costs more than ending T_i and starting T_j independently is dominated and removed; the remaining outgoing degree is bounded before solving the integer program.

The method is offline in a precise sense: it compares stable fragments on both sides of a gap and chooses all compatible continuations jointly. It does not infer through arbitrarily long absence. Gaps beyond the horizon remain fragmented, and ambiguous evidence should do the same.

5.5 Dense trajectories

Linked detections remain sparse. The post-processor removes tracks that are too short or sparsely observed, performs a forward camera-compensated Kalman pass and backward Rauch–Tung–Striebel smoothing [11], and interpolates internal gaps. The dense trajectory records whether each observation was detected or synthesized, then feeds the rider-relative stabilization stage.



Figure 4: Pose-derived virtual-camera geometry. The detector box and centre (orange) describe image extent. Boom junction and mast tip (cyan) provide a rider-relative anchor (magenta) and scale cue.

6 Pose-Guided Stabilization

Tracking determines *which* surfer is present; the downstream view must decide where to place a virtual camera and how tightly to crop. An axis-aligned detector box is a poor control signal because a tall, rotating rig changes box centre and height even when the athlete moves little. Direct box-based cropping therefore produces lateral drift and visible zoom pumping.

6.1 Two task-specific landmarks

The pose model predicts the boom–mast junction and mast tip. These points were chosen for rendering geometry rather than as a reduced human skeleton. The boom junction provides a repeatable reference near the rider; the boom-to-tip distance provides a rig-relative scale cue that is less sensitive to in-plane rotation than box height. The checked-in development dataset contains 1,123 manually labelled training images and 56 validation images. The split is image-level and can contain frames from the same source sequence on both sides, so it does not establish cross-sequence generalization.

For box top y_1 , height h , and centre (c_x, c_y) , the fallback anchor is

$$\mathbf{a}_{\text{box}} = (c_x, y_1 + 0.6h). \quad (4)$$

When the boom junction $\mathbf{b} = (b_x, b_y)$ is visible, a low-

noise mast proxy $\tilde{\mathbf{t}} = (b_x, y_1)$ gives

$$\mathbf{a}_{\text{pose}} = 0.15\tilde{\mathbf{t}} + 0.85\mathbf{b}. \quad (5)$$

The measured mast tip is intentionally excluded from position: its long lever arm amplifies localization noise. It remains useful for scale.

Small crops do not support equally reliable landmarks. The unsmoothed anchor blends pose and box geometry according to the shorter box dimension, with pose weight rising from zero at 45 pixels to one at 55 pixels. Missing boom observations are interpolated for short gaps; longer gaps ease toward the per-frame box fall-back and back to pose when reliable evidence returns. A five-frame robust weighted mean suppresses isolated jumps while preserving response to actual motion.

6.2 Rig-relative scale

When both landmarks are visible, scale begins with $\ell = \|\mathbf{b} - \mathbf{t}\|_2$. Short gaps are interpolated or held; otherwise the anchor-to-box-top distance supplies a fallback. A centred temporal mean produces $\bar{\ell}$, and the requested crop height is

$$C = \max\left(\frac{\bar{\ell}}{0.75}, \frac{h}{0.9}, 1\right), \quad s = \text{clip}\left(\frac{C}{H}, 10^{-6}, 1\right), \quad (6)$$

where H is source height. The mast segment is targeted to 75% of the output, while the box term prevents excessively tight crops. The resulting anchor and scale are stored per frame with the identity track. Global background stabilization remains a separate signal: it improves motion estimation and optional overview rendering, but cannot replace rider-relative composition when the rig rotates within the scene.

7 Evaluation

We evaluate identity association on 21 manually curated development videos. The benchmark fixes the universe of 41,079 saved, non-interpolated detections and gives every method the same bounding boxes. Of these, 41,004 belong to retained surfer identities and 75 are annotated for rejection. This deliberately isolates association from detector recall: the results answer whether the available observations were grouped correctly, not whether YOLO found every surfer.

The saved observations were selected during manual trajectory reconstruction and their boxes were Rauch–Tung–Striebel smoothed afterwards. The benchmark is therefore conditional on gold-derived observations rather than an end-to-end MOT test. It is also an in-sample development-set evaluation: algorithm design was informed by these videos, and the provenance of the original ILP parameter tuning is not fully recorded. We report the results as engineering evidence and failure analysis, not as a held-out leaderboard.

7.1 Metrics

For retained detections, pairwise precision and recall compare every unordered pair. A pair is positive when

both observations have the same annotated identity:

$$P_{\text{pair}} = \frac{\text{correct same-identity pairs}}{\text{all pairs joined by the tracker}}, \quad (7)$$

$$R_{\text{pair}} = \frac{\text{correct same-identity pairs}}{\text{all annotated same-identity pairs}}. \quad (8)$$

False merges reduce precision and fragmentation reduces recall. Missing outputs are treated as singleton predictions rather than being silently omitted. Micro scores pool all pairs and consequently weight long identities quadratically; macro scores average per-video values. We additionally report contaminated predicted tracks, which contain more than one gold identity; fragmentation excess, the number of predicted pieces beyond one per gold identity; and exact videos, whose complete partition matches the annotation up to track-label permutation. These absolute counts matter because a high aggregate F1 can conceal one application-breaking identity switch.

7.2 Fixed-observation comparison

Table 1 compares the production pipeline with OC-SORT and BoT-SORT. OC-SORT receives the fixed detections directly. BoT-SORT uses its ECC camera compensation but no pedestrian-specific ReID network. The production method reruns conservative preprocessing, masked video stabilization, and offline ILP association. Its sail descriptor uses border-derived foreground masking and a whole-fragment appearance prototype; motion gating evaluates centre, width, and height; and implausible motion, area, or aspect-ratio transitions are excluded before optimization.

The production system gives the strongest overall association score and reduces fragmentation by a factor of approximately six relative to BoT-SORT and by a factor of twenty relative to OC-SORT. Eleven of 21 videos are exact, compared with four for BoT-SORT and one for OC-SORT. OC-SORT attains the highest pairwise precision by remaining highly conservative, but its 845 excess fragments do not provide the continuous identity track required by the application. The production result is a more useful operating point for this dataset, but its nine contaminated tracks mean that it still does not satisfy the nominal zero-false-merge requirement.

Runtime is included only as an operational diagnostic. On the evaluation machine, OC-SORT took 9.0 seconds, BoT-SORT 497.7 seconds, and the production association path 402.3 seconds. The workloads are not identical: OC-SORT needs no video decode, BoT-SORT includes decode and ECC, and production includes masked stabilization but excludes the initial crop-embedding extraction.

7.3 Safeguard ablation

The initial production reconstruction exposed a concentrated false-link problem. Preprocessor fragments were almost always identity-pure, but the ILP sometimes joined two pure fragments from different surfers. Many

Table 1: Fixed-observation identity-association results on 21 development videos. “Contam.” counts predicted tracks containing more than one gold identity.

Method	Coverage	Pair P	Pair R	Pair F1	Macro F1	Contam.	Frag. excess
OC-SORT	0.981	0.985	0.662	0.792	0.857	9/165	845
BoT-SORT, no ReID	0.996	0.920	0.752	0.828	0.858	16/142	244
Production offline ILP	0.999	0.957	0.918	0.937	0.970	9/95	42

Table 2: Staged production-pipeline revision on the same development set.

Stage	P	R	F1	Cont.	Frag.
Original	0.907	0.880	0.893	20	48
+ mask, mean, 4D	0.939	0.917	0.928	17	44
+ conservative vetoes	0.957	0.918	0.937	9	42

of the unexpected short-gap errors occurred at crossings, where the endpoint crops contained both sails. The previous appearance model emphasized precisely those boundary frames, the color descriptor’s intended foreground mask was bypassed by an early return, and the ILP requested a position-only projection from a Kalman model that already represented width and height.

Table 2 records two staged revisions. Because several changes are bundled in the first stage, this is a system ablation rather than a claim that any single component produced the full difference. Foreground masking, whole-fragment appearance aggregation, and four-dimensional motion evaluation raise micro F1 from 0.893 to 0.928. Adding conservative vetoes for motion NLL and threefold area or aspect-ratio discontinuities raises precision further and halves contaminated tracks, without increasing fragmentation.

The vetoes are not a second tracker layered on top of the Kalman filter. The Kalman state already contains centre, width, and height; the earlier ILP call discarded the size dimensions. The revised system uses the full four-dimensional Mahalanobis evidence and additionally refuses links whose absolute evidence is implausible. This second step is necessary because the ILP’s start and end penalties can otherwise make even a very poor link cheaper than splitting two central fragments.

A manual audit found nine wrong selected fragment joins. Unlike the removed geometric failures, they concentrate at crossings, visually near-identical sails, and crops containing both competitors. Their implications and the unsuccessful global-margin abstention experiment are discussed in Section 8.

8 Discussion

8.1 What the comparison establishes

The benchmark supports a narrow systems claim. On the fixed observations in this development set, the offline pipeline occupies a substantially better continuity–contamination operating point than the evaluated OC-SORT and BoT-SORT configurations. This is useful because the final product requires long identity paths

rather than a large collection of locally plausible fragments. It is not a claim of general MOT superiority: the observation set is domain-specific, the evaluation is in-sample, and detector recall is outside the protocol.

The results also make the asymmetric objective concrete. OC-SORT has the highest pair precision, yet its 845 excess fragments make it operationally unsuitable. The production system reduces that count to 42, but the remaining nine contaminated tracks are still serious errors. Aggregate F1 is therefore a diagnostic rather than an acceptance criterion; final trajectories remain subject to review.

8.2 Residual failure mode

The original failure audit separated two kinds of false link. Geometrically implausible joins resulted from bypassed foreground masking, endpoint-biased appearance aggregation, and position-only use of a size-aware motion model. The revised evidence and conservative vetoes remove many of these cases. The remaining wrong joins cluster at crossings, visually near-identical sails, and crops containing both competitors. In those cases, time, position, scale, and color can all support the wrong continuation.

This residual mode is not well addressed by another global scalar threshold. An abstention experiment removed each chosen edge in turn and compared the alternative objective. The resulting margin did not separate wrong from correct links: useful thresholds rejected substantially more correct associations than errors. A credible next step would instead model the interaction explicitly—for example, by detecting paired crossings and comparing pre- and post-overlap appearance jointly. We leave that extension unimplemented rather than tuning a development-set-specific margin.

8.3 Scope and limitations

The manually reconstructed videos are the development set, not a held-out test set. The current tracker was frozen for the reported comparison, but the provenance of the older numerical ILP tuning is incomplete. Moreover, saved observations were selected during manual reconstruction and their boxes were smoothed using the gold trajectories. The study consequently evaluates association conditional on a useful set of boxes; it does not measure end-to-end detector performance and should not be read as a MOTA or HOTA result.

The baseline configurations are also intentionally pragmatic. BoT-SORT retains ECC camera compensation but omits its pedestrian ReID model, whose learned

representation does not match windsurf sails. A dedicated learned embedding might improve that baseline, but training one would duplicate the domain adaptation that the compact color descriptor is meant to avoid. The pose model is documented as part of the system and its geometry is visible in the produced crops, but it has no independent cross-video quantitative benchmark. Its image-level validation split may contain nearby frames from the same source video.

Finally, the three-second candidate horizon encodes a practical trade-off. It limits false long-range joins and keeps the graph tractable, but a genuine disappearance longer than that remains split. The system therefore approaches its application through conservative automation plus review, not through a guarantee of perfect identity recovery.

9 Conclusion

We presented an offline system that turns long-shot windsurfing video into stable, rider-relative trajectories. Its design follows the structure of the problem: masked camera-motion estimation handles large pans, conservative local association protects tracklet purity, sail-specific appearance and four-dimensional motion score longer continuations, a global integer program uses complete-video context, and two rig landmarks control the final virtual camera.

On 21 manually reconstructed development videos, the association subsystem improves pairwise F1 and greatly reduces fragmentation relative to the evaluated OC-SORT and BoT-SORT configurations. The staged revision also shows why implementation details matter: activating the intended foreground mask, aggregating appearance over whole fragments, retaining box-size evidence, and vetoing implausible links collectively reduce contaminated tracks from 20 to nine. The remaining crossing errors and the in-sample protocol preclude a claim of perfect or general-purpose tracking. The contribution is instead a reproducible account of how domain structure, offline inference, and an explicit failure cost can make an otherwise ordinary detection model useful in a demanding video-analysis system.

A Reconstruction Details

A.1 Evaluation reconstruction

The benchmark uses the 21 manually reconstructed development videos checked into the project’s evaluation manifest. It contains 41,079 non-interpolated saved observations: 41,004 assigned to 88 retained identities and 75 assigned to the discard class. Saved boxes were RTS-smoothed after manual identity reconstruction. For the production method, the historical intermediate tracklets were unavailable; the evaluator therefore starts from singleton saved observations, reruns the current preprocessor, and then runs the current ILP tracker. A missing method output is scored as a unique singleton rather than omitted from the pair metrics.

The comparison is reproduced from the repository root with:

```
python -m video_processing.inference.  
optimization.compare_trackers --no-resume
```

Machine-readable aggregate and per-video results are stored under `documentation/tracker_evaluation/`. The BoxMOT distribution used for the run is 13.0.17; its package-level version string reports 13.0.16.

A.2 Production configuration

The reported production run uses a maximum candidate gap of three seconds and at most ten outgoing candidate links per fragment. The ILP start and end costs are both 96.0993. Motion, appearance, and gap costs have weights 3.4694, 2.2492, and 9.9926; the per-frame miss probability is 0.8037 and the appearance calibration parameter is 11.6301. Motion compares the first two detections of a successor using the complete (c_x, c_y, w, h) measurement. Candidate links are rejected above motion NLL 8 or above an area or aspect-ratio change factor of 3.

The appearance prototype retains the full fragment. Tracklets with at most nine detections may be discarded, with first-detection cost 36.4497 and growth factor 1.6423. These values are reported for reconstruction, not as independently validated universal hyperparameters.

A.3 Baseline configuration

OC-SORT uses the frozen BoxMOT defaults recorded by the evaluator: confidence thresholds 0.1 and 0.2, maximum age 30, minimum hits 3, IoU threshold 0.3, $\Delta t = 3$, inertia 0.2, BYTE association disabled, $Q_{xy} = 0.01$, and $Q_s = 0.0001$. BoT-SORT uses ECC global-motion compensation with ReID disabled; its high, low, and new-track thresholds are 0.5, 0.1, and 0.6, its buffer is 30, its match threshold is 0.8, and score fusion is disabled. All methods receive the same fixed observations.

References

- [1] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, “Simple Online and Realtime Tracking,” *ICIP*, 2016.
- [2] N. Wojke, A. Bewley, and D. Paulus, “Simple Online and Realtime Tracking with a Deep Association Metric,” *ICIP*, 2017.
- [3] Y. Zhang et al., “ByteTrack: Multi-Object Tracking by Associating Every Detection Box,” *ECCV*, 2022.
- [4] J. Cao, J. Pang, X. Weng, R. Khrodar, and K. Kitani, “Observation-Centric SORT: Rethinking SORT for Robust Multi-Object Tracking,” *CVPR*, 2023.
- [5] N. Aharon, R. Orfaig, and B.-Z. Bobrovsky, “BoT-SORT: Robust Associations Multi-Pedestrian Tracking,” *arXiv:2206.14651*, 2022.
- [6] G. Maggolino, A. Ahmad, J. Cao, and K. Kitani, “Deep OC-SORT: Multi-Pedestrian Tracking by Adaptive Re-Identification,” *ICIP*, 2023.
- [7] L. Zhang, Y. Li, and R. Nevatia, “Global Data Association for Multi-Object Tracking Using Network Flows,” *CVPR*, 2008.
- [8] S. Li, Y. Kong, and H. Rezatofighi, “Learning of Global Objective for Network Flow in Multi-Object Tracking,” *CVPR*, 2022.
- [9] J. Luiten et al., “HOTA: A Higher Order Metric for Evaluating Multi-Object Tracking,” *IJCV*, 2021.
- [10] R. E. Kalman, “A New Approach to Linear Filtering and Prediction Problems,” *Journal of Basic Engineering*, 1960.
- [11] H. E. Rauch, F. Tung, and C. T. Striebel, “Maximum Likelihood Estimates of Linear Dynamic Systems,” *AIAA Journal*, 1965.